SMART INDIA HACKATHON 2026

Problem Statement ID - SIH26175

Problem Statement Title - Single-View Height Estimation and 3D Flythrough

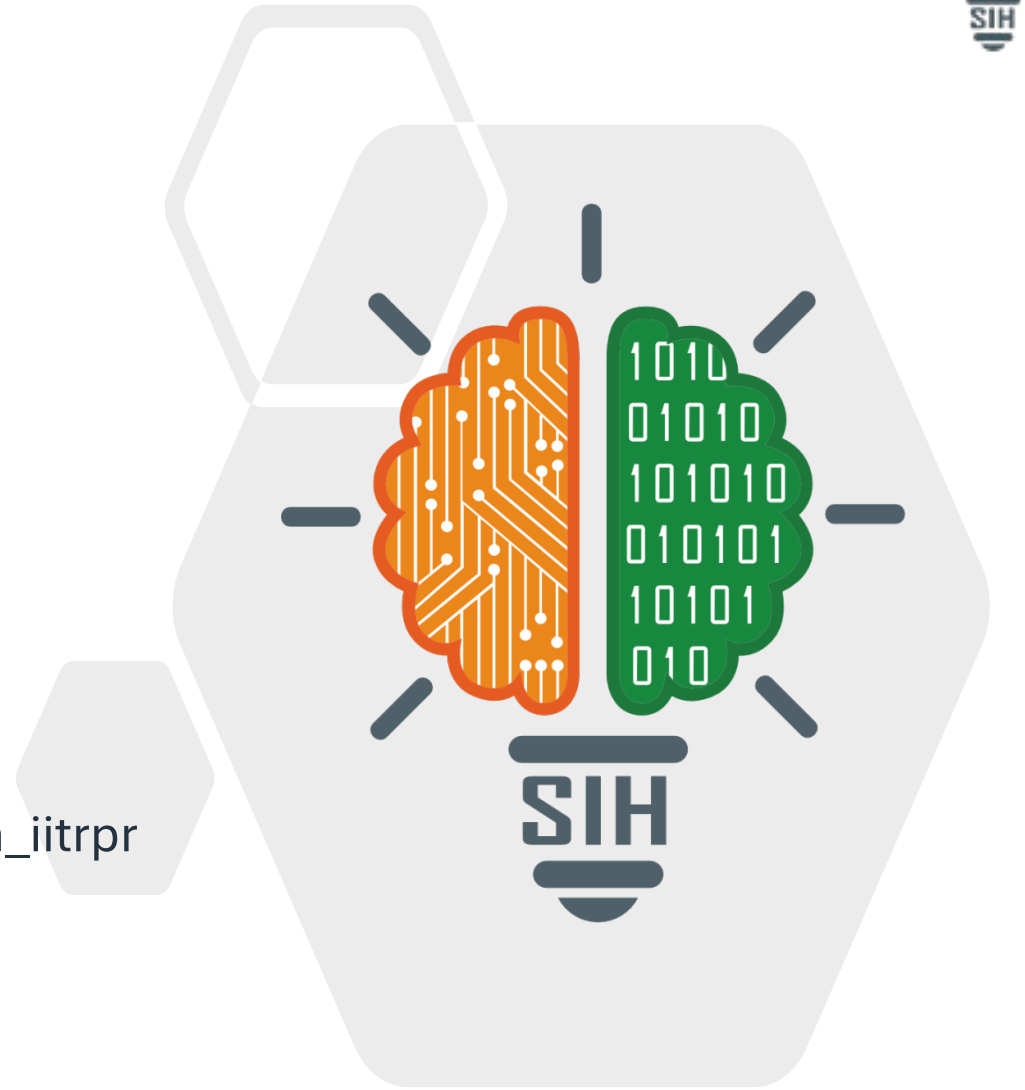
Theme - Disaster Management

PS Category – Software

Team ID - [Registered Team ID]

Team Name (Registered on portal) - TerraVista_iitrpr

Idea Title - DepthWizard



IDEA TITLE

DepthWizard: One Satellite Image → Measured 3D Heights You Can Fly Through and Verify

INPUT	AI HEIGHT	CALIBRATE	VERIFY	FLY & ACT
PNG / JPG / TIFF / GeoTIFF	Height above ground, every pixel	Terrain DEM + ICESat-2 → real metres	ICESat-2 laser check + click-to-measure	3D flythrough + flood / slope tools

HOW IT MEETS THE PROBLEM STATEMENT

- ✓ One optical image in: no stereo pair, LiDAR or InSAR needed
- ✓ PNG/JPG → relative DSM · GeoTIFF → absolute DSM in metres
- ✓ Upload → auto-process → fly through the reconstructed 3D terrain
- ✓ Click any building: estimated height ± uncertainty, roof level in m
- ✓ Working prototype: runs offline on a laptop; GeoTIFF + GLB export

WHY DEPTHWIZARD IS DIFFERENT

- Satellite-trained height model; generic depth AI fails on top-down images
- Terrain levelled with NASA ICESat-2 laser: flood map CSI 0.30 → 0.79
- Trust layer: per-pixel uncertainty (± flags where the error is high)
- Bird's-eye → street-level view in one click (Sat3DGen mesh next)
- Disaster tools on top: flood depth, landslide slopes, affected buildings

INNOVATION & UNIQUENESS

Satellite-native height AI	**Real metres, laser-levelled**	**Laser-verified (ICESat-2)**	**Bird's-eye → street flythrough**

TECHNICAL APPROACH

Pipeline: Pixels → Height → Metres → Verified 3D

INPUT — RGB image (PNG / JPG / TIFF)

PREPROCESS — Read GSD + CRS · overlapping 518 px tiles

AI MODEL — RS3DAda (ViT-L + DPT) · challenger: DINOv3-SAT

HEIGHT — Height above ground + uncertainty map

CALIBRATE — GLO-30 / CartoDEM terrain, buildings removed → metres

VERIFY — ICESat-2 correction + error report

OUT — DSM + 3D

AI / ML

- ✓ RS3DAda (NeurIPS 2024, MIT) runs on an 8 GB laptop
- • Challenger: DINOv3-SAT + DPT (CHMv2 init), GAMUS-tuned
- ✓ Measured on GAMUS + DFC2019 test tiles (MAE / RMSE)
- ✓ 4-flip TTA → uncertainty map · blended 518 px tiles
- • Distilled small model → ONNX Runtime on CPU

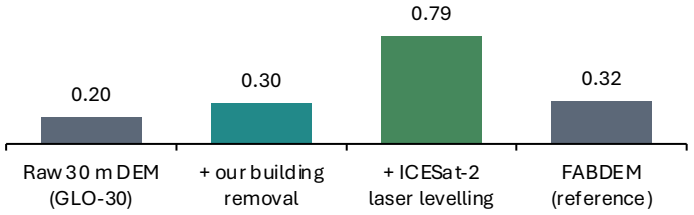
GEOSPATIAL

- ✓ Rasterio · PyProj; GeoTIFF float32 in / out
- ✓ DSM = terrain (GLO-30; CartoDEM drop-in) + height
- • Prompt2DEM (2025): 30 m DEM + RGB → 30 cm heights
- ✓ ICESat-2 ATL03 / ATL08: level terrain + report error
- ✓ One vertical datum (EGM2008; ICESat-2 converted)

APPLICATION & 3D VIEWER

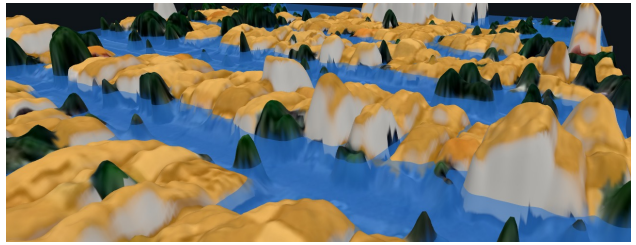
- ✓ Three.js viewer; GLB + GeoTIFF export
- ✓ Free-fly (WASD), draw-a-path auto-fly, MP4 export
- • Street mode: Sat3DGen (ICLR 2026) street mesh
- ✓ Click-to-measure; flood + landslide-slope tools
- ✓ Runs offline on a laptop · • Tauri desktop app

MEASURED: Porto Alegre 2024 flood (CSI)



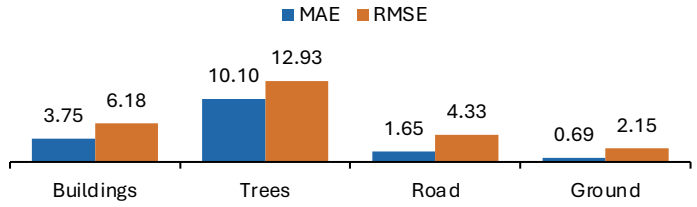
Our run: flood predicted from pre-flood images in 4 test areas vs the IPH-UFRGS LiDAR flood map. CSI 1 = perfect

PROTOTYPE: flood fly-through, saved as MP4



Our viewer: drone path over Canoas (Brazil) at the May 2024 flood level; orange = roofs still above water

MEASURED on GAMUS, untuned model (m)



Our run: released RS3DAda, 150 GAMUS test tiles vs LiDAR; official MAE 3.88 / RMSE 5.56 m. Trees: next fix (GAMUS fine-tuning)

✓ working in our prototype today • planned (TARGET) | built for both halves of the score: DSM accuracy + Rendering & UX

FEASIBILITY AND VIABILITY

Built on Released Models, Free Data and Offline-Ready Software

FEASIBILITY

- Working baseline on day 1: RS3DAda weights + inference script are public (MIT)
- Training data ready: GAMUS (PS-recommended), DFC2019/US3D, SynRS3D (69k synthetic images)
- Free terrain + truth data: CartoDEM (Bhuvan), SRTM, Copernicus GLO-30, ICESat-2
- Fine-tuning fits one cloud GPU; frozen-backbone option for low budgets

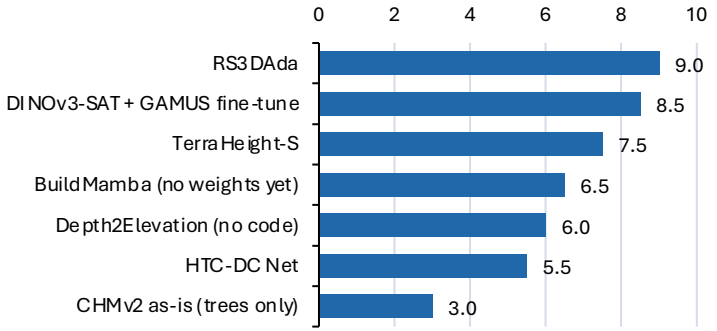
VIABILITY

- Modular: swap the height model without touching calibration or the viewer
- Standard outputs (GeoTIFF DSM, GLB) open in QGIS, ArcGIS and Bhuvan workflows
- Offline today on a laptop; next: ONNX CPU model + Tauri desktop app
- License-clean: MIT / Apache / DINOv3 license; no non-commercial weights

CHALLENGE → RISK → MITIGATION

CHALLENGE	RISK	MITIGATION
New image resolution (GSD)	Wrong height scale	Read GSD from metadata · resample · multi-resolution fine-tuning
One image can't see ground level	Heights off in metres	Remove buildings from terrain · ICESat-2 laser levelling (built)
Few tall buildings in training data	Tall buildings under-estimated	Oversample tall buildings · report error by height band
Tilted (off-nadir) images	Leaning buildings, shifted roofs	Off-nadir training data (US3D) · warning in UI
No GPU / no internet in the field	Tool unusable	Offline laptop app (built) · ONNX CPU model next

Model shortlist (team assessment, /10)



Score = fit to PS + public weights + reported accuracy + effort. Multi-view models (need several images) excluded.

IMPACT AND BENEFITS

From One Image to Faster, Verifiable Disaster Decisions

DISASTER MANAGEMENT

Flood-depth simulation · landslide-prone slopes · affected-building counts
Tested on the 2024 Porto Alegre flood: overlap with the LiDAR flood map, CSI 0.79

DEPTHWIZARD

Measured 3D from one image

FROM TERRAIN DATA TO ACTIONABLE INTELLIGENCE

URBAN PLANNING

Building heights and volumes · 3D base maps where no LiDAR exists

REMOTE SENSING (ISRO)

More value from existing optical archives · fresh 3D from every new image, no stereo tasking

1 **HEIGHT MAP** per pixel

2 **VERIFIED 3D** metres + ICESat-2

3 **DECISION** flood / slope

One optical image → measured, checked terrain context

FIELD RESPONSE

Fly-through + street view for team briefings · line-of-sight / viewshed for relief staging

ECONOMIC

Reuse existing images instead of new LiDAR/stereo surveys for every event

SOCIAL

Faster 3D damage picture; works offline in the field

ENVIRONMENTAL

Terrain, slope and canopy context for flood, landslide and forest monitoring

RESEARCH AND REFERENCES

Research Foundation: 2024–2026 Models, Open Data, Independent Validation

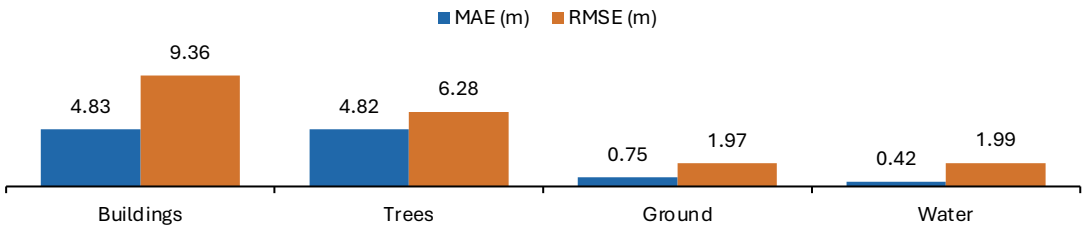
MODELS & METHODS

- RS3DAda / SynRS3D: Song et al., NeurIPS 2024 (primary model)
- DINOv3 (Meta, 2025) + CHMv2 (Scientific Data, 2026): satellite backbone
- Prompt2DEM: Rafaeli et al., 2025, SRTM-prior metric heights (30 m → 30 cm)
- ICESat-2-guided correction: Song, Chen & Yokoya, ISPRS J. 2026
- Sat3DGen (ICLR 2026) · Benchmarks: BuildMamba 2026, Depth2Elevation (TGRS 2025), HTC-DC Net

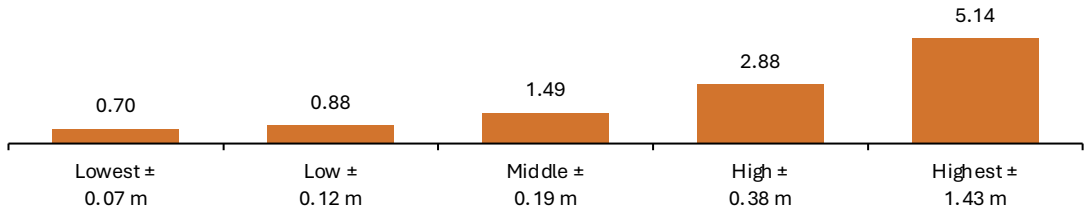
DATASETS / VALIDATION / OFFICIAL SOURCES

- GAMUS (recommended in PS) · DFC2019 / US3D · SynRS3D
- CartoDEM (ISRO Bhuvan) · SRTM 30 m · Copernicus DEM GLO-30
- ICESat-2 ATL03 / ATL08 (NASA) · IPH-UFRGS 2024 flood map
- ISRO Bhoonidhi / Bhuvan imagery · ISPRS Potsdam / Vaihingen
- Metrics: RMSE and MAE per terrain type and per height band

MEASURED: RS3DAda error vs LiDAR by land cover (m)



MEASURED: where our ± is high, real error is high (MAE, m)



Same 200 test tiles; pixels split into 5 equal groups by predicted ± (spread of 4 flipped predictions). Rank correlation 0.62.

Our run: released RS3DAda, 200 DFC2019 test tiles (Jacksonville + Omaha). Official metric MAE 2.22 m / RMSE 3.83 m.

MODELS	RS3DAda + DINOv3	**DATASETS**	GAMUS + SynRS3D	**TERRAIN REFERENCE**	CartoDEM / SRTM / GLO-30	**VALIDATION**	RMSE / MAE + ICESat-2

Research basis: monocular height estimation • DSM generation • geospatial calibration • ICESat-2 validation • terrain rendering | Evaluation: RMSE / MAE